

Philmont Scout Ranch Geology for Advanced Hikers

Executive Summary

Philmont is geologically interesting because it compresses a very long history into a relatively small hiking landscape: ancient Precambrian crystalline basement, Pennsylvanian redbeds, Jurassic and Cretaceous sedimentary cover, coal-bearing latest Cretaceous to Paleocene strata of the Raton Basin, Paleogene intrusive rocks that now define landmarks such as Baldy-area ridges and the Tooth of Time, and younger basalt related to the Ocate Mesa–Raton–Clayton volcanic province just south and east of the core ranch. In practical field terms, Philmont is a place where hikers can walk from marine shale and shoreline sandstone into dinosaur-bearing and coal-bearing strata, then into steeply tilted redbeds, and finally into Precambrian metamorphic and plutonic rocks within a single trek. ¹

USGS Professional Paper 505 remains highly useful for field interpretation because it is built around traverses that still match how hikers actually experience the ranch; its strongest contribution is spatial reasoning about the “rock pile,” unconformities, intrusive sheets, and structure. Where it is dated is mainly in chronology and regional interpretation: it uses broad relative-age assignments for many intrusive and volcanic rocks, predates the discovery and formal description of the Philmont tyrannosaur track, and predates modern regional syntheses that frame the Cimarron Range as a Laramide uplift and distinguish older Philmont intrusions from the younger Raton–Clayton volcanic field. ²

For advanced hikers, the core interpretive idea is this: Philmont’s landscape is an erosional cross section through an east-flank mountain uplift and adjacent basin margin. Lower-country trails typically cross younger Cretaceous to Paleocene basin strata such as the Pierre, Trinidad, Vermejo, Raton, and Poison Canyon units; the central and northwestern country exposes older Dakota, Morrison, Entrada, Dockum, Sangre de Cristo-equivalent redbeds, and then crystalline basement; landmarks such as the Tooth of Time and parts of the Baldy country owe their relief to intrusive rocks that resisted erosion better than surrounding sedimentary units. ³

Mining in the Baldy district was not a random historical accident; it reflects structurally controlled mineralization associated with fractures, faults, and favorable contacts. At least for the French Henry mine, USGS MRDS records gold, silver, and copper commodities; chalcopyrite, chrysocolla, malachite, and quartz gangue; and ore localized in faults, fractures, and along shale-sandstone contacts. Pettit’s New Mexico Geological Society summary ties the district’s development to placer discoveries on Baldy’s flanks and to rich sandstone-hosted veins at the Aztec mine. ⁴

The trek-specific implication for Trek 12–15 is that you should expect mostly basin-margin clastic rocks and coal-bearing units in the east and north, progressively older and more steeply tilted sedimentary rocks around the mountain front and Baldy country, obvious mining modifications around Baldy Town and adjacent drainages, and a return across lower-country erosional surfaces near Ponil and Indian Writings.

Your itinerary includes Harlan, Devils Wash Basin, Cimarron River, Santa Claus, Baldy Town, Pueblano, Elkhorn, Ponil, Indian Writings, and Six Mile Gate return. [filecite turn0file0](#)

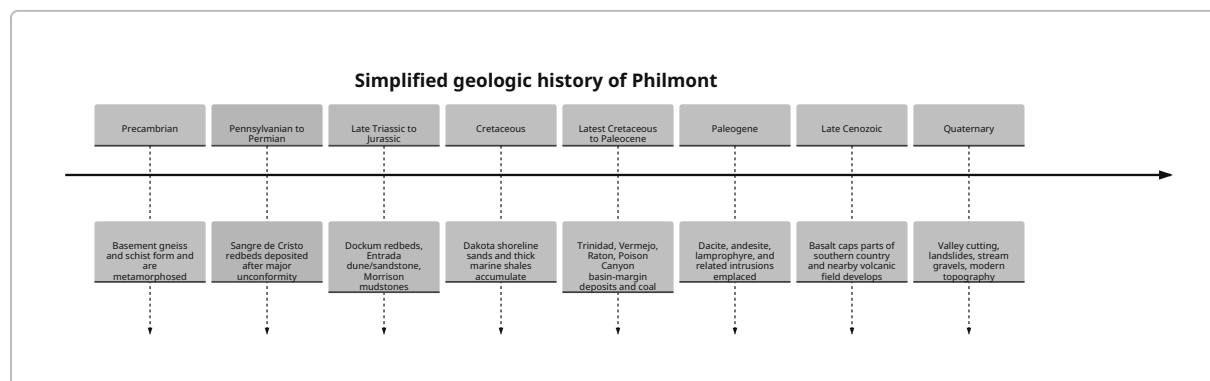
Regional Setting and Geologic History

Modern summaries describe the Cimarron Range as a Laramide uplift on the east margin of the Southern Rocky Mountains, bounded by a west-side normal fault and an east-side thrust fault. Philmont occupies a key position where that uplift meets the western edge of the Raton Basin; that geometry explains why the ranch exposes both mountain-core crystalline rocks and younger basin-fill strata in such close proximity.

5

PP 505's field traverses show the same structural story from the ground: repeated east- or northeast-dipping successions, steepening near the mountain front, older units reappearing upstream and upslope, and local removal of formations by erosion before younger units were laid down. One of the clearest examples is the unconformity beneath the Poison Canyon and Raton interval near Cimarron, where the break cuts down from Vermejo through Trinidad and ultimately onto Pierre within only a few miles. [6](#)

The long geologic sequence is, in broad terms, as follows. First came formation and metamorphism of basement rocks now exposed as gneiss and schist, which PP 505 already inferred to be Precambrian. Over that basement, redbeds of the Sangre de Cristo interval accumulated after an enormous time gap. Younger Mesozoic redbeds and sandstones followed, including Dockum-equivalent redbeds, Entrada Sandstone, Morrison Formation, Dakota Sandstone, and thick Cretaceous shales and shoreline to coastal-plain deposits. Latest Cretaceous to Paleocene time is represented by the Trinidad, Vermejo, Raton, and Poison Canyon succession that records shoreline retreat, coal-forming swamps, fluvial deposition, and basin reorganization around the Raton Basin. Much later, intrusive magmatism emplaced dacite, andesite, lamprophyre, and related bodies; still later basalt flowed across the southern country. [7](#)



A crucial interpretive refinement beyond PP 505 is that newer regional work places Philmont's basaltic surroundings within the broader Raton-Clayton volcanic field and the Jemez Lineament system, whereas PP 505 could only assign the basalt a broad late Tertiary age. That does **not** mean Baldy or the Tooth of Time are part of the young Raton-Clayton field; rather, the accessible evidence points to older Paleogene intrusive rocks for the iconic Philmont monoliths and younger late Cenozoic surface volcanism for the basaltic mesas to the south and east. [8](#)

Stratigraphic Framework and Field Recognition

The table below is the highest-confidence field summary for major Philmont units and features, synthesized from PP 505's rock-sequence traverses, regional unit summaries, and newer public references where available. Exact unit boundaries on any one trail are local-map problems; this table is meant to be a field booklet guide, not a substitute for a full geologic map. ⁹

Formation or feature	Approximate age	Brief description and lithology	Typical field appearance	Where to expect it on Philmont	Map reference or public example	Suggested field observation or question
Precambrian basement	Precambrian	Gneiss, mica schist, hornblende schist, quartzite, granodiorite, diorite	Foliated, crystalline, sparkly mica, banding; no sedimentary bedding	Mountain core, especially around high western and southwestern country	PP 505 notes gneiss and schist beyond upper canyon and near Beaubien/ Fowler country ¹⁰	Are you seeing foliation and mineral segregation rather than bedding?
Sangre de Cristo redbeds	Pennsylvanian–early Permian	Red to gray sandstone, conglomerate, mudstone	Thick red-gray clastic beds; coarse where proximal	Near mountain-front older sedimentary package, especially west/ southwest sectors	PP 505 unit 9 red-gray sandstone and conglomerate; regional Sangre de Cristo summary ¹¹	Do clast size and bedding suggest nearby highlands shedding debris?
Dockum Group redbeds	Late Triassic	Red shale, siltstone, sandstone	Slope-forming red beds, softer than Dakota/ Entrada	Upper canyon to Cimarroncito/ Rayado side of the ranch	PP 505 unit 8 Dockum Group ¹²	Are mudstone slopes hiding most of the unit except sparse ledges?
Entrada Sandstone	Jurassic	Quartz sandstone	Light-colored, cleaner sandstone ledges, thinner than Dakota in PP 505	Upper Cimarron and related western canyon sections	PP 505 unit 7 Entrada Sandstone near Gravel Pit Lakes/ Palisades sector ¹³	Cross-beds or clean quartzose texture present?

Formation or feature	Approximate age	Brief description and lithology	Typical field appearance	Where to expect it on Philmont	Map reference or public example	Suggested field observation or question
Morrison Formation	Late Jurassic	Red shale and sandstone	Red slope-forming unit between more resistant sandstones	Cimarroncito, Fowler/ Rayado traverses	PP 505 unit 6 Morrison Formation ¹⁴	Does it form a recessive interval between stronger ledges above and below?
Dakota Sandstone	Cretaceous	Yellow quartz sandstone, locally in two ledges with shale between	Resistant yellowish quartzose ledges, cliffy ridges	Ute Park, upper Cimarron, Fowler/ Rayado sections	PP 505 unit 5 Dakota Sandstone ¹⁵	Is this the most obvious cliff-former in your immediate section?
Graneros and Pierre shale package	Cretaceous	Thick black shale; locally limestone in PP 505 section	Dark, slope-forming shale; poor exposure; occasional limestone ledges	Lower-country plains and canyon bottoms; mountain-front black-shale belt	PP 505 unit 4 as Pierre in lower Cimarron and Graneros in upper canyon; limestone near Stone Wall Pass area ¹⁶	Is vegetation or landslide cover masking the shale more than the rock itself?
Trinidad Sandstone	Latest Cretaceous	Fine arkosic shoreline sandstone	Light-gray to buff sandstone cliff or ledge	Near Cimarron and lower benchlands	PP 505 unit 3 Trinidad; modern unit summary calls it final seaway-regression shore deposit ¹⁷	Does the unit look beach/shoreface-like compared with coal-bearing beds above?
Vermejo Formation	Latest Cretaceous	Sandstone, shale, coal	Alternating gray-yellow sandstone and carbonaceous beds; coal dumps where mined	Northern/ lower benchlands near Cimarron corridor	PP 505 unit 2 Vermejo; modern unit summary emphasizes coal-bearing nature ¹⁸	Can you spot black carbonaceous beds or human spoil from coal?

Formation or feature	Approximate age	Brief description and lithology	Typical field appearance	Where to expect it on Philmont	Map reference or public example	Suggested field observation or question
Raton and Poison Canyon formations	Latest Cretaceous–Paleocene	Fluvial sandstone, conglomerate, shale, coal in Raton; coarser arkosic fluvial clastics in Poison Canyon	Yellow-brown sandstone/ conglomerate benches, broad erosional surfaces	Lower and northern country benchlands	PP 505 unit 1 and unconformity discussion; modern summaries distinguish gray coal-bearing Raton from yellower Poison Canyon ¹⁹	Are you on a broad bench cut on resistant fluvial sandstone rather than a sharp ridge?
Dacite porphyry intrusive sheets, plugs, monoliths	Paleogene or broad Tertiary in PP 505	Intermediate intrusive rock forming sheets, ledges, monoliths	Light-colored, blocky, resistant, commonly with feldspar phenocrysts	Tooth of Time, Palisades, Baldy-area intrusive country, many ridges	PP 505 treats dacite as younger than Poison Canyon and older than Quaternary gravel; public summaries describe Tooth as dacite intrusion/ laccolithic body ²⁰	Are you looking at a body that cuts bedding rather than follows it?
Andesite, lamprophyre, dark diorite, basalt	Cenozoic	Younger igneous bodies and basalt caps	Darker, denser, blockier rocks; basalt capping mesas	Baldy north-side intrusive country; Fowler Mesa/ Ocate margin for basalt	PP 505 relative-age discussion and basalt cap at Fowler Mesa ²¹	Does the basalt sit flat across older tilted rocks, proving it is younger?

Igneous, Structural, and Landscape Interpretation

The most useful structural model for the ranch is that Philmont exposes a tilted, eroded layer cake wrapped around a mountain-core uplift. PP 505 repeatedly shows units reappearing across multiple drainages, identifies the range as a broad arch, and demonstrates that many streams cut across structure rather than simply following it. Modern regional summaries refine that by describing the Cimarron Range as a Laramide uplift with a west-side normal fault and east-side thrust fault. ²²

Field-scale structure matters constantly on Philmont. The Poison Canyon/Raton unconformity near Cimarron cuts progressively downward from Vermejo to Trinidad to Pierre in only a short distance, which is why lower-country hikers can see abrupt changes in cliff character and bench-forming behavior. Near the mountain front, units stand much more steeply, so apparent outcrop width can be misleading; PP 505 explicitly warns that some units only look thinner because they are nearly vertical. ²³

The igneous story is equally central. Dacite porphyry occurs both as intrusive sheets and as larger masses; PP 505 emphasizes that these bodies were “squeezed between layers as a melt,” so they cannot be treated as simple surface beds. The Palisades along upper Cimarron are a classic resistant intrusive expression. In the south and southwest country, Tooth of Time and related landmarks are best interpreted as erosional remnants of intrusive bodies, not youthful volcanic cones. ²⁴

For Baldy versus Tooth of Time, the safest synthesis is this. Both are linked to Paleogene intrusive magmatism in the broader Philmont region, but they are geomorphically different. Baldy is a high topographic culmination within the structurally uplifted and intruded northern mountain block; the Tooth is a more isolated monolithic erosional remnant that owes its prominence to contrast in resistance between intrusion and host rock. Public summaries place the Tooth within a dacite intrusive or laccolithic setting, while PP 505 establishes the regional abundance of dacite sheets and related intrusive bodies around the mountain front and upper canyons. ²⁵

The late Cenozoic basalt story is different. PP 505 recognized basalt capping Fowler Mesa and Ocate-country surfaces and correctly interpreted it as younger than the tilted sedimentary and crystalline rocks beneath. Modern regional work goes much further by tying nearby basaltic volcanism to the Raton-Clayton volcanic field and Jemez Lineament, with much younger ages than the older Philmont intrusive suite. That is one of the clearest places where modern interpretation substantially refines PP 505. ²⁶

Mining Geology and Fossils

The Baldy district geology makes sense as a coupled lode-and-placer system. Pettit describes how placer gold discoveries on the flanks and drainages of Baldy led prospectors upslope to hard-rock sources, especially the Aztec mine, where he reports rich production from veins in sandstone. The district’s hydraulic and placer history also explains why hikers still encounter disturbed gulches, ditch remnants, altered slopes, and anthropogenic benching around the Baldy country. ²⁷

French Henry is the best-documented individual mine in the material retrieved here. The USGS MRDS entry records gold and silver as primary commodities, copper as tertiary, quartz gangue, and ore minerals including chalcopyrite, chrysocolla, and malachite. Crucially, it states that mineralization occurs in faults, fractures, and along the shale-sandstone contact. The same record gives a public WGS84 example coordinate for the main adit at about 36.64032, -105.19139 and identifies five or six adits. That is exactly the kind of structurally controlled, contact-focused ore localization hikers should look for conceptually, even where access to actual workings is restricted. ²⁸

Historically visible mining features most worth interpreting are therefore these: waste rock and tailings that mark underground workings rather than natural scree; tramway or mill-site placements that reveal gravity-fed ore handling; altered or iron-stained wall rock near contacts and fractures; and placer-modified stream bottoms downstream from lode sources. At French Henry specifically, public descriptions note an

abandoned restricted mine area upslope and a mill site with surviving cabins and interpretive use near the South Ponil. 29

Philmont’s single most famous fossil claim is the large tyrannosaur footprint from North Ponil Canyon. Public summaries report discovery by Charles Pillmore in 1983, formal description by Lockley and Hunt in 1994, an age near the latest Cretaceous, and a footprint about 83 cm long by 70 cm wide. Even if one prefers the more cautious wording “probable tyrannosaur” rather than “definitive T. rex,” the track is unquestionably one of Philmont’s major post-PP 505 paleontological additions, because PP 505 predates its discovery by almost two decades. 30

The broader fossil story relevant to hikers is narrower than the ranch’s rock diversity might suggest. Philmont proper is better known publicly for trace fossils and age-diagnostic fossils used in stratigraphy than for abundant roadside body fossil collecting. In other words, the rocks are fossil-important, but most of the value to hikers lies in reading environments—marine shale, shoreline sandstone, coal swamp, fluvial channel, dune or sheet sand—not in expecting museum-style fossil abundance on the trail. PP 505 itself notes that most named formations in or near Philmont contain enough fossils to establish geologic age, even though the booklet does not function as a collecting guide. 31

Trek 12–15 Route Interpretation

Your Trek 12–15 itinerary runs through Harlan, Devils Wash Basin, Cimarron River, Santa Claus, Baldy Town, Pueblano, Elkhorn, Ponil, Indian Writings, and the Six Mile Gate return. That implies a route crossing lower-country basin strata, entering the older and more structurally complex Baldy country, then returning through northern-country erosion surfaces and drainages. 32

The table below is therefore deliberately approximate. It matches the itinerary to the **most likely** major formations and rock environments crossed, based on PP 505 traverses and the known geography of the camps, not on a turn-by-turn GPS field map. 32

Formation name	Age	Lithology	Expected outcrop appearance	Significance to hikers
Poison Canyon Formation	Paleocene	Coarse arkosic sandstone, conglomerate, mudstone	Yellow-brown bench-forming sandstone and broad slopes	Explains much of the lower-country bench topography and easier walking surfaces near the Cimarron corridor
Raton Formation	Latest Cretaceous–Paleocene	Sandstone, conglomerate, shale, coal	Gray to buff clastic beds with carbonaceous intervals	Good place to think about coal swamps, K–Pg time, and basin-margin rivers
Vermejo Formation	Latest Cretaceous	Sandstone, shale, coal	Mixed sandstone/shale with local coal evidence and dumps	Human mining history and carbonaceous beds become legible here

Formation name	Age	Lithology	Expected outcrop appearance	Significance to hikers
Trinidad Sandstone	Latest Cretaceous	Fine arkosic sandstone	Distinct ledges and cliff bands	Classic shoreline marker unit; helps hikers visualize retreat of the Western Interior Seaway
Pierre and Graneros shale package	Cretaceous	Black shale with local limestone	Vegetated dark slopes, poor exposure, occasional ledges	Usually the reason trail tread is smoother or more soil-covered than the dramatic cliffs above and below
Dakota Sandstone	Cretaceous	Quartz sandstone	Bold yellow quartzose ledges and ridges	One of the easiest units to recognize; a reliable "hard ledge" on approach to older country
Morrison Formation	Late Jurassic	Red shale and sandstone	Recessive red slopes between stronger sandstone units	Useful marker when counting upward or downward in the section
Entrada Sandstone	Jurassic	Quartz sandstone	Lighter clean sandstone ledge, thinner than Dakota	Signals entry into older Mesozoic section approaching the mountain front
Dockum Group redbeds	Late Triassic	Red mudstone, shale, sandstone	Softer red slopes with sparse ledges	Important reminder that Philmont includes pre-Jurassic continental redbeds, not just Cretaceous basin rocks
Sangre de Cristo redbeds	Pennsylvanian–early Permian	Red-gray sandstone and conglomerate	Coarse thicker red-gray clastics	A strong cue that you are entering the oldest preserved sedimentary cover around Baldy country
Dacite porphyry intrusions	Paleogene	Porphyritic intrusive rock	Light, blocky, resistant cliffs, monoliths, sills	Key to understanding why Baldy-country ridges and iconic pinnacles stand up so strongly

Formation name	Age	Lithology	Expected outcrop appearance	Significance to hikers
Precambrian basement	Precambrian	Gneiss, schist, granodiorite, diorite	Crystalline, foliated, nonbedded rock	The moment when hikers leave the sedimentary story and enter mountain-core crust

A practical route-reading heuristic is useful here. Around Harlan, Devils Wash Basin, Cimarron River, Ponil, and Indian Writings, think first in terms of Raton Basin stratigraphy, erosional benches, shale slopes, and coal-bearing or fluvial clastic rocks. Around Santa Claus, Baldy Town, Pueblano, and Elkhorn, think increasingly in terms of older tilted strata, intrusive bodies, contact relationships, and mining-modified terrain. [filecite68turn0file068](#) ³³

Revisions, Map Products, and Limitations

The most important refinements to PP 505 are straightforward. First, modern work frames the Cimarron Range more explicitly as a Laramide uplift with specific bounding structures, whereas PP 505 was strongest at local structure but more restrained in regional tectonic wording. Second, late Cenozoic basaltic volcanism is now interpreted within the Raton-Clayton volcanic field and Jemez Lineament context, rather than simply as broadly “late Tertiary” lava. Third, the famous North Ponil tyrannosaur track simply did not exist in the Philmont literature when PP 505 was published. ³⁴

PP 505 is **not** obsolete. Its route-based geologic reasoning, unconformity interpretation, and unit-by-unit hill-and-canyon observations remain highly valuable for anyone actually walking Philmont. What has changed is the precision of age assignments, the regional tectonic framework, and the paleontological inventory. For a field booklet, the best approach is to use PP 505 as the spatial backbone and modern sources as calibration. ³⁵

Primary-source links and identifiers most worth building into a custom booklet are listed below.

USGS Professional Paper 505 landing page:

https://pubs.usgs.gov/pp/pp_505/

USGS PP 505 full PDF:

https://pubs.usgs.gov/pp/pp_505/pdf/pp_505.pdf

NMGS Pettit, History of mining in Colfax County, New Mexico:

https://nmgs.nmt.edu/publications/guidebooks/downloads/17/17_p0069_p0075.pdf

DOI: 10.56577/FFC-17.69

USGS MRDS French Henry Mine:

https://mrdata.usgs.gov/mrds/show-mrds.php?dep_id=10012974

USGS Professional Paper 68, The ore deposits of New Mexico:
DOI: 10.3133/pp68

Open questions remain. I was able to assemble high-confidence field interpretation from USGS and NMGS sources, but I was **not** able in this session to retrieve every newer map or paper that would allow a perfect modern unit-by-unit replacement for PP 505, especially for exact radiometric ages of individual Philmont dacite/andesite bodies and for a fully current map-sheet correlation of every trail segment. So the report is strongest on field relationships, stratigraphic order, mining controls, and route interpretation, and more cautious on exact intrusion ages and some finer paleontological details. ³⁶

1 2 3 9 11 12 13 14 15 16 17 18 19 22 24 26 32 33 35 GSPP 505

https://pubs.usgs.gov/pp/pp_505/pdf/pp_505_j.pdf

4 28 29 French Henry Mine (MRDS #10012974) AU, AG

https://mrdata.usgs.gov/mrds/show-mrds.php?dep_id=10012974

5 34 Cimarron Range, New Mexico

https://en.wikipedia.org/wiki/Cimarron_Range%2C_New_Mexico?utm_source=chatgpt.com

6 7 8 10 20 21 23 31 36 GSPP 505

https://pubs.usgs.gov/pp/pp_505/pdf/pp_505_k.pdf

25 Tooth of Time

https://en.wikipedia.org/wiki/Tooth_of_Time?utm_source=chatgpt.com

27 History of mining in Colfax County, New Mexico

https://nmgs.nmt.edu/publications/guidebooks/downloads/17/17_p0069_p0075.pdf

30 Tyrannosauripus - Wikipedia

<https://en.wikipedia.org/wiki/Tyrannosauripus>